Summer Physics Research Internship

University of Hawaii at Manoa

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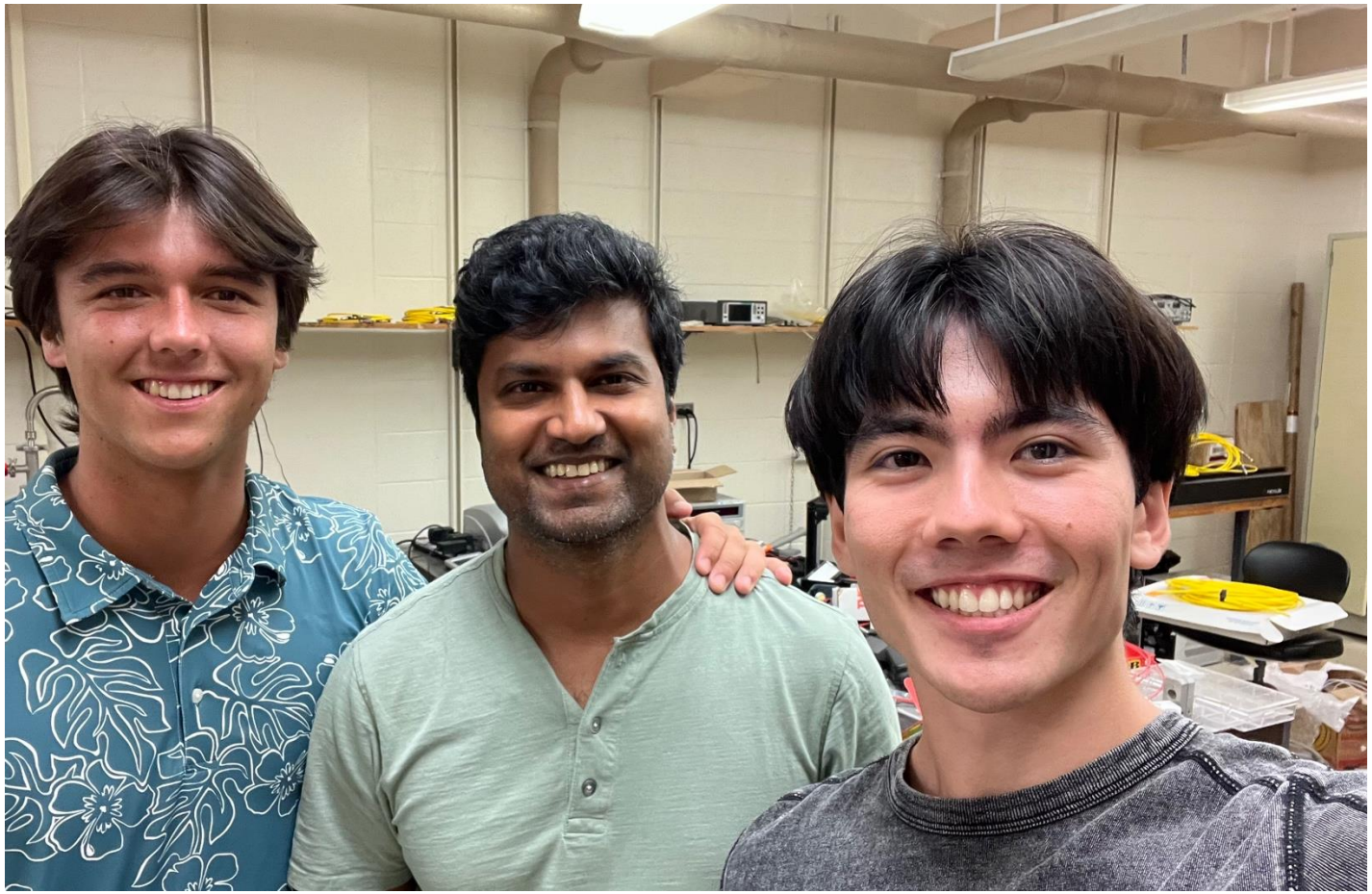
Overview

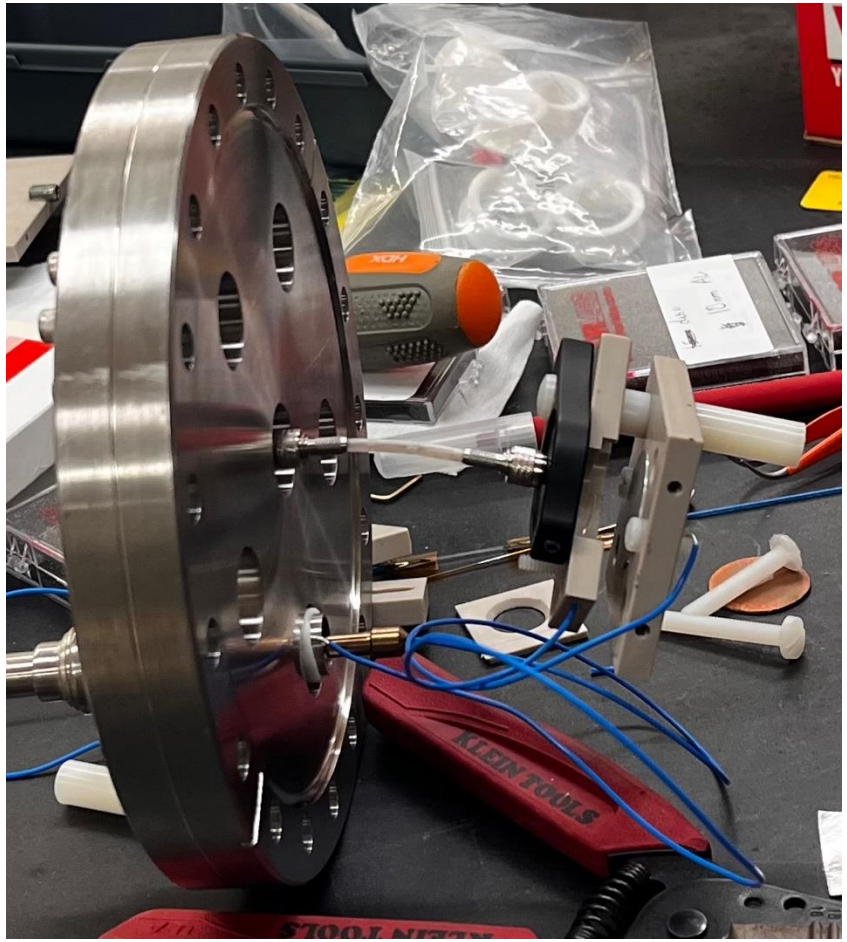
This past summer, I was involved in experimental physics work regarding the quantum efficiencies of photocathode setups. This was in effort to create a prototype for the liquid-argon-submerged photocathode setup at DUNE.

We tested both transmissive and reflective setups in a vacuum chamber operating at ~1.0 hPa and a 10 Hz pulse laser operating at ~8.0 µJ through fiber-optic wire to simulate photomultiplier output in liquid argon. Liquid nitrogen was used for temperature testing.

I worked under Ranjan Dharmapalan and with Evan Porter in Jelena Maricic's lab at UH Manoa.

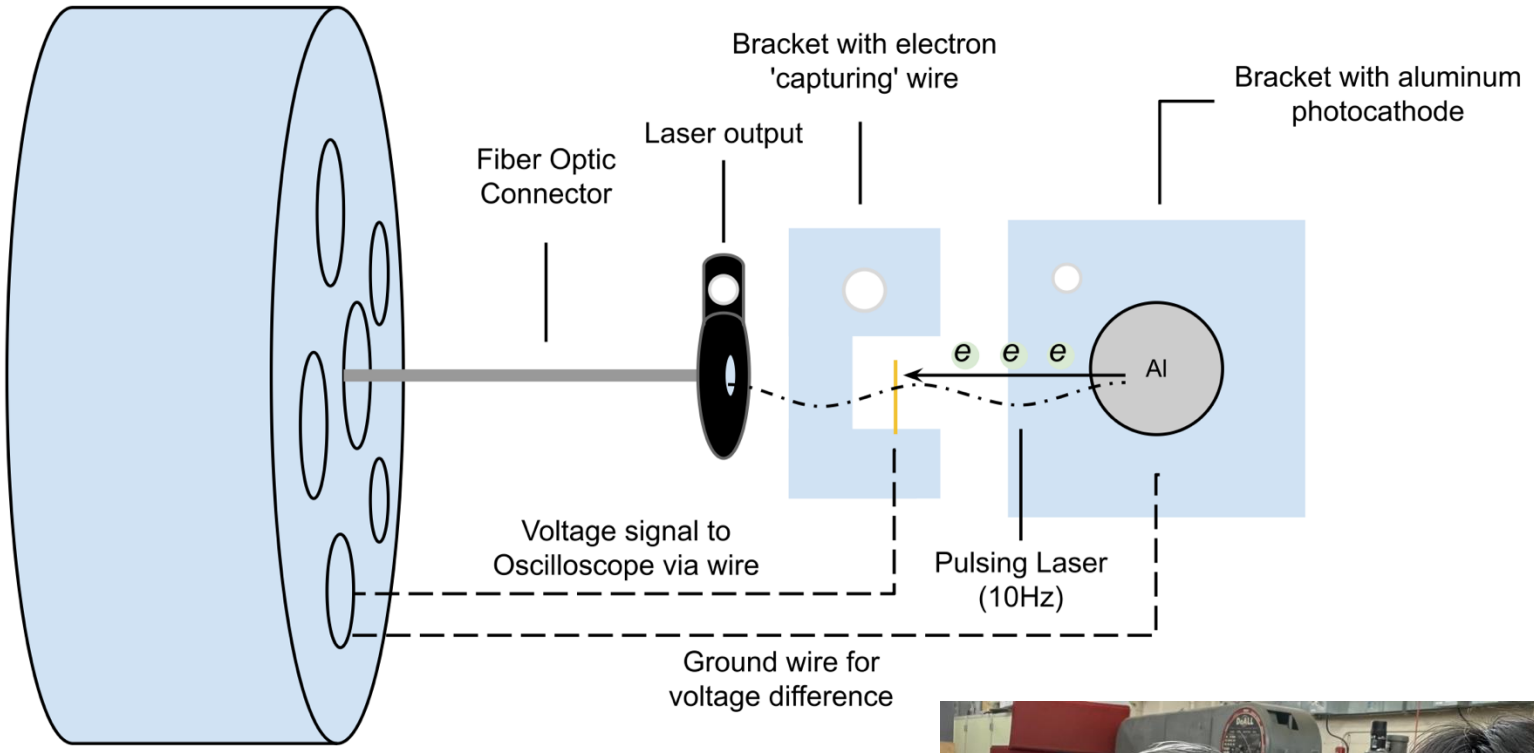


Selfie with colleagues: Former classmate Evan Porter (left), Postdoc supervisor Ranjan Dharmapalan (middle)



Vacuum Chamber Internals

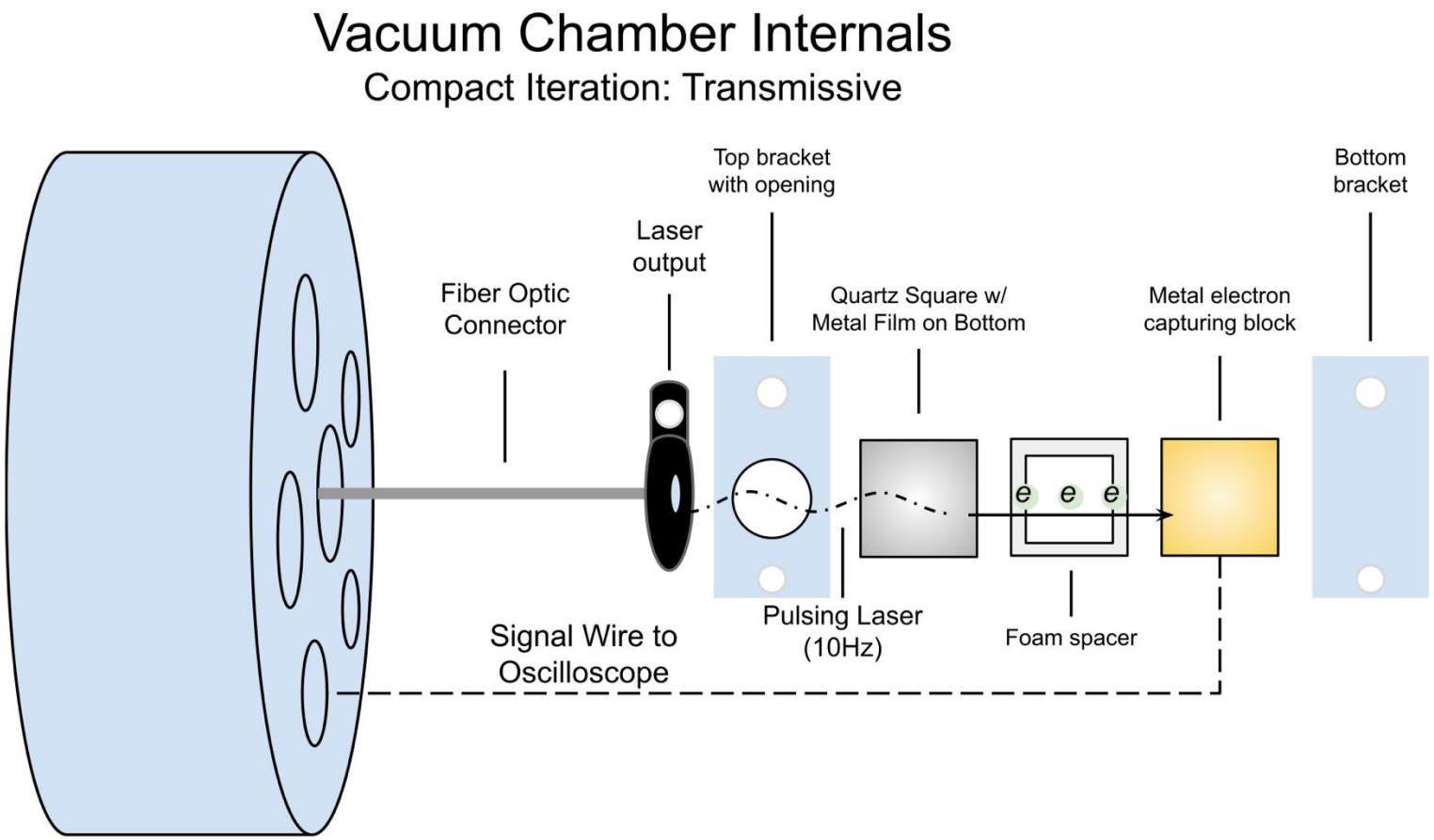
Early Iteration: Reflective



Our reflective setups featured either an aluminum disc or a quartz square with a metallic film. In an early setup shown above, a wire above the photocathode would transmit the photoelectric signal out of the vacuum chamber.

We created and modified necessary components in the machine shop under instruction of Roy Tom, operating a drill press and other machinery.

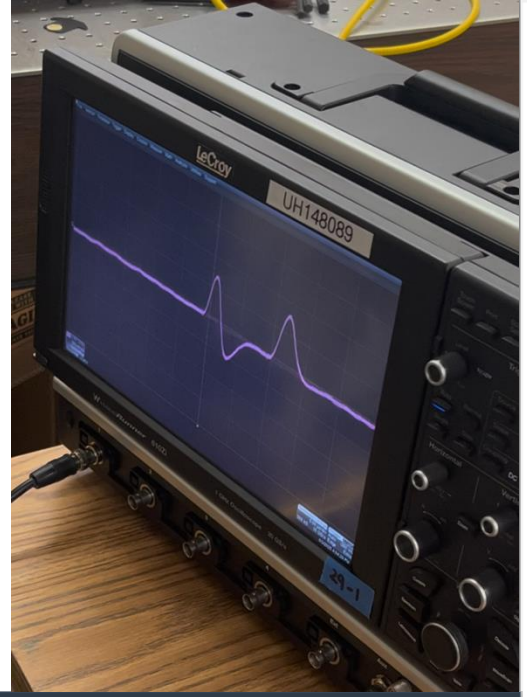
Working with machinist Roy Tom (left)

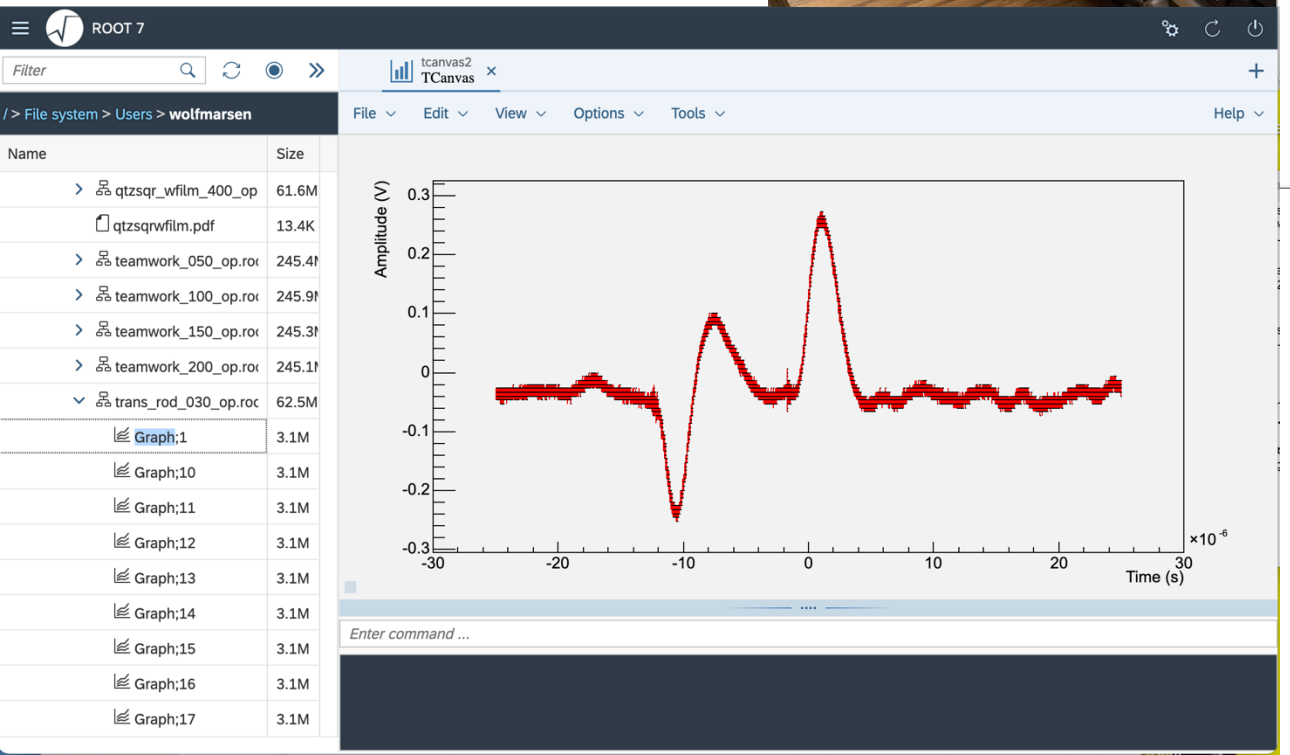


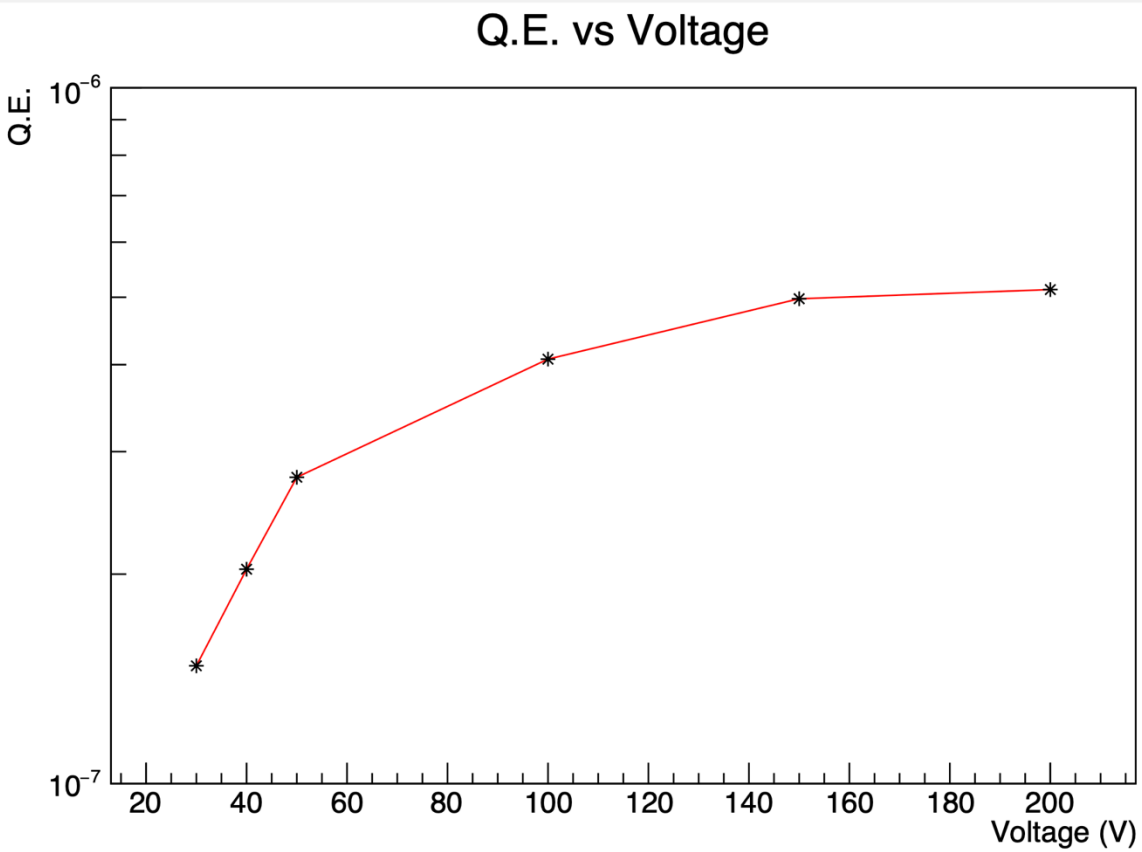
Our transmissive setups featured primarily a quartz square with a metallic film. The compact setup shown in the picture and diagram above has a wire connected to a metal block for more electron capture below a quartz square to transmit the photoelectric signal.

We used an oscilloscope to interpret the pulsed signals into code-readable files.

In practice, we took 10-20 pulsed voltage readings and took the average of the maximums.







We then used C++ and CERN's ROOT framework to find the quantum efficiency of a setup at varying voltages and graph data.

Using drill press to fabricate G-10 plate and filling liquid nitrogen tank to test photocathode adhesive thermal retention